Lab: Installing MySQL Server on Ubuntu (Linux)

To Begin with the Lab

Summary

In this lab, we will learn how to install, configure, connect to, and uninstall the MySQL Server on an Ubuntu Linux distribution.

We will use Ubuntu terminal commands to install MySQL Server, configure the root user, and connect to the database using MySQL Workbench.

We will also learn how to stop the MySQL service and completely remove MySQL packages, dependencies, and configuration files from Ubuntu.

Understanding MySQL Server

MySQL Server is an open-source relational database management system (RDBMS) used to store, manage, and retrieve data.

It uses SQL (Structured Query Language) to perform database operations such as:

- Creating databases and tables.
- Inserting records.
- Updating existing records.
- Deleting records.
- Retrieving data.

Why do we install MySQL Server on Ubuntu?

MySQL Server can run directly on a Linux-based machine. Developers can use it to manage databases locally without depending on a remote database server.

Example:

Suppose we are developing an employee management application.

We can create a MySQL database named EmployeeDB and store employee information.

EmployeeID	Name	Department
1	Rahul	IT
2	Priya	HR
3	Amit	Cloud

The application can connect to MySQL Server and perform database operations.

Lab Steps

Step 1: Update Ubuntu Packages

Before installing MySQL Server, update the Ubuntu package repository.

Command:

```
sudo apt update
```

```
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$ sudo apt update
[sudo: authenticate] Password:
Hit:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://security.ubuntu.com/ubuntu resolute-security InRelease
Hit:3 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:4 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
72 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

Explanation:

- sudo: Executes the command with administrator privileges.
- apt: Ubuntu package management tool.
- update: Updates the package information.

Step 2: Install MySQL Server

Execute the following command:

```
sudo apt install mysql-server -y
```

```
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$ sudo apt install mysql-server -y
Installing:
  mysql-server

Installing dependencies:
  libaio1t64             libfcgi-bin              libhtml-tagset-perl    libio-html-perl         libtcmalloc-minimal4t64  mysql-client-core
  libcgi-fast-perl       libfcgi-perl             libhtml-template-perl  liblwp-mediatypes-perl  libtimedate-perl         mysql-common
  libcgi-pm-perl         libfcgi0t64              libhttp-date-perl      libmecab2               libunwind8               mysql-server-core
  libclone-perl          libgoogle-perftools4t64  libhttp-message-perl   libnuma1                liburi-perl
  libencode-locale-perl  libhtml-parser-perl      libicu78               libprotobuf-lite32t64   mysql-client

Suggested packages:
  libdata-dump-perl  libipc-sharedcache-perl  libio-compress-brotli-perl  libbusiness-isbn-perl  libmime-base32-perl  libregexp-ipv6-perl  libwww-perl

Recommended packages:
  mecab-ipadic-utf

Summary:
  Upgrading: 0, Installing: 29, Removing: 0, Not Upgrading: 72
  Download size: 36.3 MB
  Space needed: 223 MB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 mysql-common all 5.8+1.1.1ubuntu2 [7002 B]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libtcmalloc-minimal4t64 amd64 2.18.1-1 [138 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libunwind8 amd64 1.8.3-0ubuntu1 [59.6 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute/main amd64 libgoogle-perftools4t64 amd64 2.18.1-1 [199 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 mysql-client-core amd64 8.4.11-0ubuntu0.26.04.1 [2200 kB]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 mysql-client amd64 8.4.11-0ubuntu0.26.04.1 [9126 B]
Get:7 http://archive.ubuntu.com/ubuntu resolute/main amd64 libaio1t64 amd64 0.3.113-8build1 [7356 B]
Get:8 http://archive.ubuntu.com/ubuntu resolute/main amd64 libicu78 amd64 78.2-2ubuntu1 [11.5 MB]
Get:9 http://archive.ubuntu.com/ubuntu resolute/main amd64 libmecab2 amd64 0.996-15.1build4 [205 kB]
Get:10 http://archive.ubuntu.com/ubuntu resolute/main amd64 libnuma1 amd64 2.0.19-1build1 [23.7 kB]
Get:11 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 libprotobuf-lite32t64 amd64 3.21.12-15ubuntu1.1 [249 kB]
Get:12 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 mysql-server-core amd64 8.4.11-0ubuntu0.26.04.1 [19.4 MB]
Get:13 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 mysql-server amd64 8.4.11-0ubuntu0.26.04.1 [1596 kB]
Get:14 http://archive.ubuntu.com/ubuntu resolute/main amd64 libhtml-tagset-perl all 3.24-1 [14.1 kB]
```

Explanation:

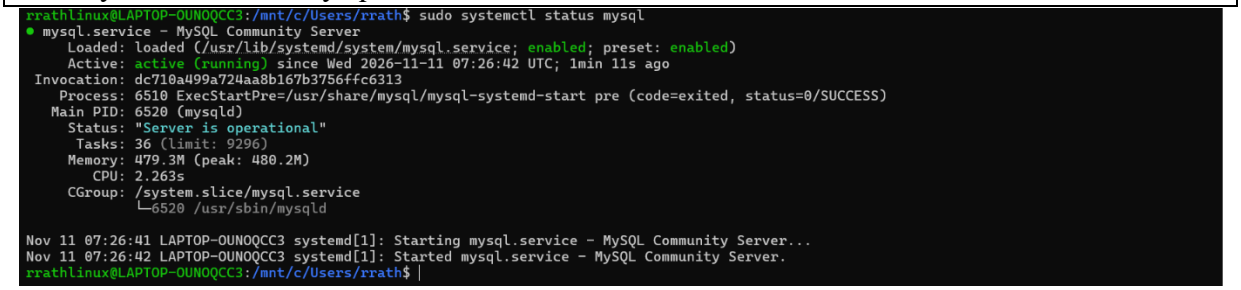
- apt install: Installs a software package.
- mysql-server: MySQL Server package.
- -y: Automatically confirms the installation.

This installs MySQL Server on Ubuntu.

Step 3: Check MySQL Service

After installation, check whether the MySQL service is running.

```
sudo systemctl status mysql
```



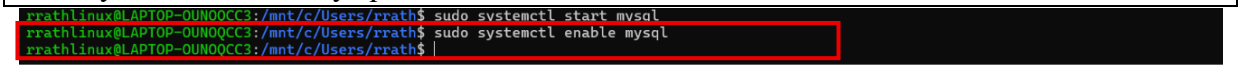
If it is not running, start it:

```
sudo systemctl start mysql
```



Enable MySQL to start automatically when the machine boots:

```
sudo systemctl enable mysql
```



Step 4: Connect to MySQL Server

To access the MySQL command-line interface, execute:

```
sudo mysql
```

This opens the MySQL shell.

You should see a prompt similar to:

mysql>

Now you can execute SQL commands.

Step 5: Configure Root User Password

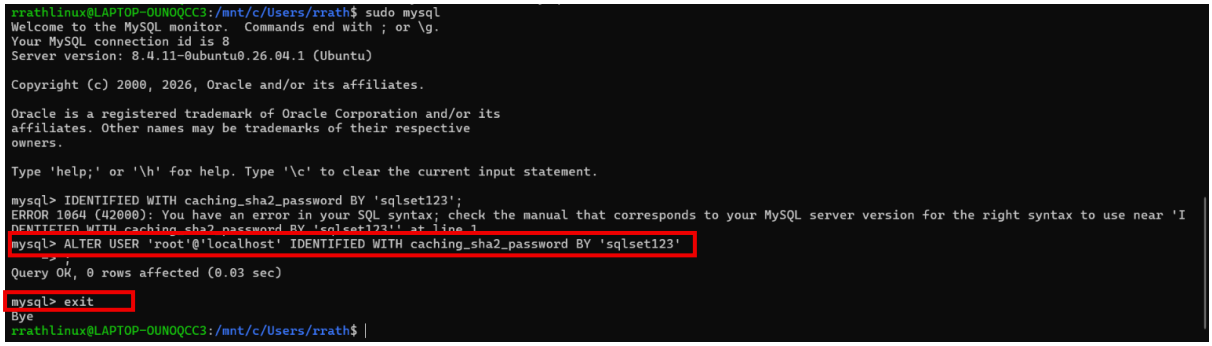
In the MySQL shell, execute:

```
ALTER USER 'root'@'localhost' IDENTIFIED WITH caching_sha2_password BY 'sqlset123' ;
```

Replace YourStrongPassword with your own strong password.

Then exit MySQL:

EXIT;



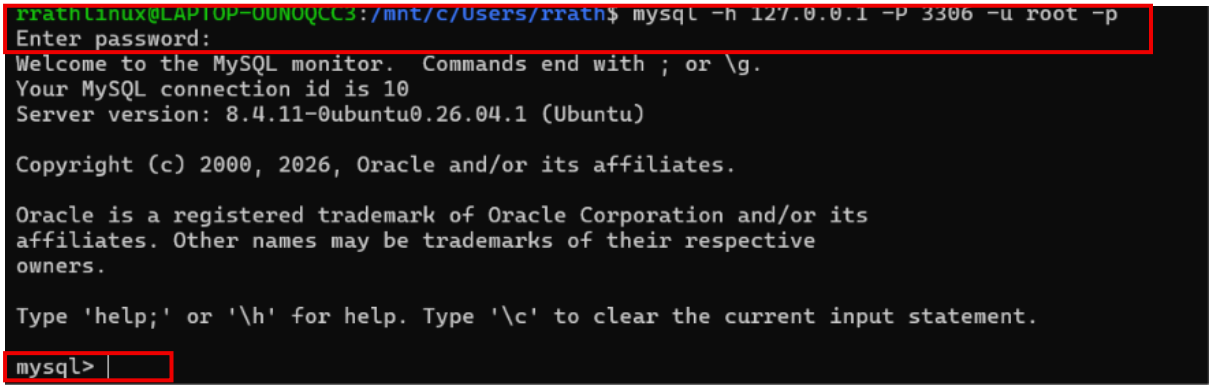
Note: Ubuntu's default MySQL installation may use the auth_socket authentication plugin for root. The above command changes root authentication to password-based authentication.

Step 6: Connect Using Password

After configuring the root password, connect using:

```
mysql -u root -p
```

Enter your configured password when prompted.



If successful, you will enter the MySQL shell.

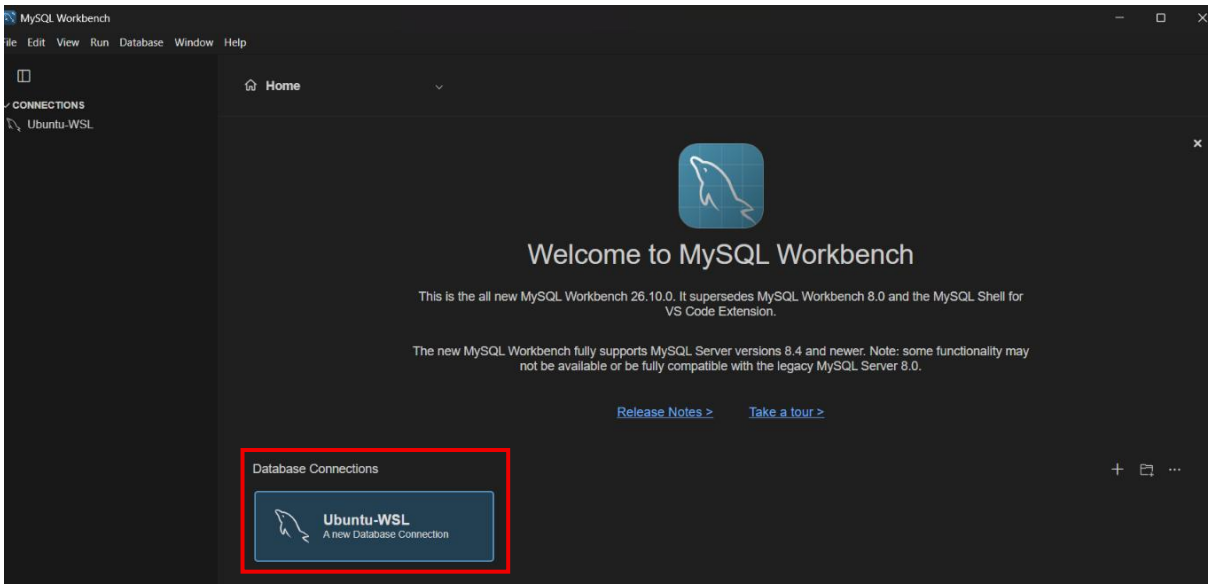
Step 7: Connect MySQL Workbench

MySQL Workbench provides a graphical user interface (GUI) to manage MySQL databases.

Steps:

1. Open MySQL Workbench.
2. Select the existing localhost connection or create a new connection.
3. Set the hostname to localhost.
4. Set the port to 3306.
5. Set the username to root.
6. Enter your configured password.
7. Click **Test Connection**.
8. Click **OK** or connect.

If authentication succeeds, you can manage your databases through MySQL Workbench.



Important: If MySQL Workbench is installed on Windows and MySQL Server is running inside WSL, localhost connectivity depends on WSL networking and port forwarding configuration.

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4. MySQL Cleanup – Uninstall MySQL Server

After completing the lab, we can remove MySQL Server from Ubuntu.

Step 1: Stop MySQL Service

Before removing MySQL, stop its service.

sudo systemctl stop mysql

This stops the MySQL Server process.

```
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$ sudo systemctl stop mysql
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$
```

Step 2: Remove MySQL Server Packages

Execute:

```
sudo apt remove mysql-server mysql-client mysql-common -y
```

This removes the specified MySQL packages.

```
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$ sudo systemctl stop mysql
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$ sudo apt remove mysql-server mysql-client mysql-common -y
The following packages were automatically installed and are no longer required:
  libaio1t64             libfcgi-bin              libhtml-tagset-perl    libio-html-perl         libtcmalloc-minimal4t64  mysql-server-core
  libcgi-fast-perl       libfcgi-perl             libhtml-template-perl  liblwp-mediatypes-perl  libtimedate-perl
  libcgi-pm-perl         libfcgi0t64              libhttp-date-perl      libmecab2               libunwind8
  libclone-perl          libgoogle-perftools4t64  libhttp-message-perl   libnuma1                liburi-perl
  libencode-locale-perl  libhtml-parser-perl      libicu78               libprotobuf-lite32t64   mysql-client-core
Use 'sudo apt autoremove' to remove them.

REMOVING:
  mysql-client  mysql-common  mysql-server

Summary:
  Upgrading: 0, Installing: 0, Removing: 3, Not Upgrading: 72
  Freed space: 1827 kB

(Reading database ... 38077 files and directories currently installed.)
Removing mysql-server (8.4.11-0ubuntu0.26.04.1) ...
update-alternatives: using /etc/mysql/my.cnf.fallback to provide /etc/mysql/my.cnf (my.cnf) in auto mode
Removing mysql-client (8.4.11-0ubuntu0.26.04.1) ...
Removing mysql-common (5.8+1.1.1ubuntu2) ...
Processing triggers for man-db (2.13.1-1build1) ...
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$
```

Step 3: Remove Unnecessary Dependencies

Execute:

```
sudo apt autoremove -y
```

This removes packages and dependencies that are no longer required.

```
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$ sudo apt autoremove -y
REMOVING:
  libaio1t64             libfcgi-bin              libhtml-tagset-perl    libio-html-perl         libtcmalloc-minimal4t64  mysql-server-core
  libcgi-fast-perl       libfcgi-perl             libhtml-template-perl  liblwp-mediatypes-perl  libtimedate-perl
  libcgi-pm-perl         libfcgi0t64              libhttp-date-perl      libmecab2               libunwind8
  libclone-perl          libgoogle-perftools4t64  libhttp-message-perl   libnuma1                liburi-perl
  libencode-locale-perl  libhtml-parser-perl      libicu78               libprotobuf-lite32t64   mysql-client-core

Summary:
  Upgrading: 0, Installing: 0, Removing: 26, Not Upgrading: 72
  Freed space: 222 MB

(Reading database ... 38036 files and directories currently installed.)
Removing mysql-server-core (8.4.11-0ubuntu0.26.04.1) ...
Removing libaio1t64:amd64 (0.3.113-8build1) ...
Removing libcgi-fast-perl (1:2.17-1) ...
Removing libhtml-template-perl (2.97-2build1) ...
Removing libcgi-pm-perl (4.71-1build1) ...
Removing libhttp-message-perl (7.01-1ubuntu1) ...
Removing libclone-perl:amd64 (0.47-1) ...
Removing libencode-locale-perl (1.05-3) ...
Removing libfcgi-bin (2.4.5-0.1build2) ...
Removing libfcgi-perl (0.82+ds-3build3) ...
Removing libfcgi0t64:amd64 (2.4.5-0.1build2) ...
Removing mysql-client-core (8.4.11-0ubuntu0.26.04.1) ...
Removing libgoogle-perftools4t64:amd64 (2.18.1-1) ...
Removing libhtml-parser-perl:amd64 (3.83-1ubuntu0.1) ...
Removing libhtml-tagset-perl (3.24-1) ...
Removing libhttp-date-perl (6.06-1ubuntu0.26.04.1) ...
Removing libicu78:amd64 (78.2-2ubuntu1) ...
Removing libio-html-perl (1.004-3) ...
Removing liblwp-mediatypes-perl (6.04-2) ...
Removing libmecab2:amd64 (0.996-15.1build4) ...
Removing libnuma1:amd64 (2.0.19-1build1) ...
Removing libprotobuf-lite32t64:amd64 (3.21.12-15ubuntu1.1) ...
Removing libtcmalloc-minimal4t64:amd64 (2.18.1-1) ...
Removing libtimedate-perl (2.3300-2) ...
Removing libunwind8:amd64 (1.8.3-0ubuntu1) ...
Removing liburi-perl (5.34-2build1) ...
Processing triggers for man-db (2.13.1-1build1) ...
Processing triggers for libc-bin (2.43-2ubuntu2.4) ...
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$
```

Step 4: Remove Package Configuration Files

Execute:

```
sudo apt purge mysql-server mysql-client mysql-common -y
```

The purge command removes package configuration files along with the packages.

```
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$ sudo apt purge mysql-server mysql-client mysql-common -y
Package 'mysql-client' is not installed, so not removed
REMOVING:
  mysql-common*  mysql-server*

Summary:
  Upgrading: 0, Installing: 0, Removing: 2, Not Upgrading: 72
  Space needed: 0 B / 1024 GB available

(Reading database ... 37492 files and directories currently installed.)
Purging configuration files for mysql-server (8.4.11-0ubuntu0.26.04.1) ...
Purging configuration files for mysql-common (5.8+1.1.1ubuntu2) ...
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$
```

Step 5: Clean Remaining MySQL Data

Warning: The following commands permanently delete MySQL databases, configuration files, and stored data.

Only execute them if you no longer need your MySQL databases.

```
sudo rm -rf /var/lib/mysql
```

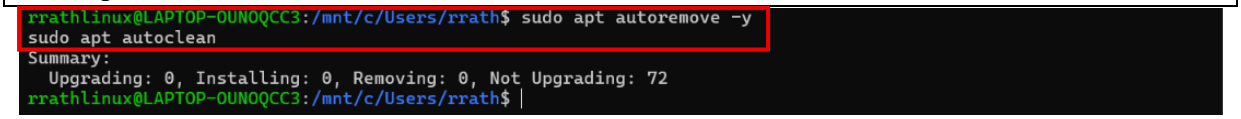
Remove MySQL configuration:

```
sudo rm -rf /etc/mysql
```



Step 6: Clean Ubuntu Package Cache

```
sudo apt autoremove -y
sudo apt autoclean
```



Step 7: Verify Removal

Execute:

```
mysql --version
```

If MySQL has been completely removed, the command should report that MySQL is not installed or that the command cannot be found.

